

OralScan AI

Agentic AI for Oral Cancer Screening & Tooth Health

口腔癌筛查与牙齿健康的多智能体 AI 系统

User Guide & Technical Report — Bilingual Edition

使用指南 与 技术报告 — 中英双语版

Live App 在线系统	https://um-idp-oral-cancer-agentic-ai.vercel.app
Architecture 架构介绍页	https://um-idp-oral-cancer-agentic-ai.vercel.app/introduction
Source 源代码	github.com/TKY0527/um-idp-oral-cancer-agentic-ai
Version 版本	June 2026 · University IDP Prototype 大学 IDP 原型

Important: Educational prototype — NOT a medical device, NOT a diagnosis. Always consult a qualified dentist or doctor.

重要提示：本系统为教育用途原型，不是医疗器械，结果不构成诊断。如有疑虑请咨询专业牙医或医生。

PART 1 • HOW TO USE — 使用指南

1. Accounts & Quick Start • 账号与快速开始

Open the live app and sign in with a one-tap demo account. Each role sees a different dashboard.
打开在线系统，点击登录页上的一键演示账号即可进入。不同角色看到不同的工作台。

Account 账号	Password 密码	Role 角色	What you see 看到什么
patient1	patient1	Patient 患者	Demo patient Aisyah (29, Malay) — profile, brushing chart, history 演示患者 Aisyah 的资料、刷牙图表与历史
patient2	patient2	Patient 患者	Demo patient Tan Wei Ming (58, smoker, high risk) 演示患者陈伟明（吸烟、高风险）
test1	test1	Patient 患者	Empty test account 空白测试账号
doctor	doctor	Doctor 医生	Triage queue, all patients, AI agents 分诊队列、全部患者、AI 智能体
admin	123456	Admin 管理员	/admin — control the 3 API keys 管理 3 个 API 密钥

2. Patient — Run a Screening • 患者端：运行筛查

- Step 1 — Provide a mouth photo: pick a sample case, upload a file, or use the camera.
第一步——提供口腔照片：选择示例病例、上传文件或直接用相机拍摄。
- Step 2 — Fill the short risk questionnaire (age, tobacco, alcohol, betel quid, lesion duration...).
第二步——填写简短风险问卷（年龄、烟酒、槟榔、病灶持续时间等）。
- Step 3 — Pick the vision AI: Mock (offline) / Gemini / Claude / ChatGPT, then press Run.
第三步——选择视觉 AI：Mock（离线）/ Gemini / Claude / ChatGPT，然后点击运行。
- Optional — 'Plug in your own API key': paste your own ChatGPT / Gemini / Claude key; it is used directly for your requests and stays in your browser only.
可选——『使用自己的 API 密钥』：粘贴你自己的 ChatGPT / Gemini / Claude 密钥，仅保存在你的浏览器中，并直接用于你的请求。

ONE photo returns TWO results: Function 1 — oral-cancer cues (visual finding, region, probability, 0–100 risk score) and Function 2 — tooth health (cavity 蛀牙 risk, gum condition, plaque, staining, hygiene score, and whether professional scaling is needed).

一张照片同时得到两类结果：功能一——口腔癌线索（视觉发现、部位、概率、0–100 风险分）；功能二——牙齿健康（蛀牙风险、牙龈状况、牙菌斑、染色、卫生评分，以及是否需要洗牙）。

On Medium/High risk a 3-expert AI panel automatically debates the case and a moderator summarises the verdict. The History page shows your risk-trend chart.

当风险为中/高时，3 位 AI 专家会自动会诊并由主持人汇总结论。『历史』页面会显示你的风险趋势图。

3. Doctor — Review & AI Agents • 医生端：审阅与 AI 智能体

- Triage Queue: every screening from web AND Telegram, sorted by urgency.
分诊队列：来自网页和 Telegram 的所有筛查，按紧急程度排序。

- Patients page: click ' Load demo patients' once — Aisyah / Tan / Muthu appear with full records (name, age, race, habits, brushing times, last scaling date, prior reports, charts, heatmap).

患者页面：点一次『加载演示患者』——Aisyah / Tan / Muthu 即出现，包含完整档案（姓名、年龄、种族、习惯、刷牙时间、上次洗牙日期、过往报告、图表与热力图）。

- AI Triage Agent (top of Patients page): ask 'Which patient should I see first?' — the agent calls its compare_all_patients tool and ranks the cohort with reasons. Watch the tool-trace chips.

AI 分诊智能体（患者页顶部）：问『我应该先看哪位病人？』——智能体调用 compare_all_patients 工具，给出带理由的排序。 标签实时显示它调用了哪些工具。

- Per-patient AI Assistant: summarise history, risk trend, red flags, scaling status — answers with real numbers fetched live by its tools.

单个患者的 AI 助手：总结病史、风险趋势、危险信号、洗牙情况——其工具实时取数，用真实数字回答。

- Session review: agree / upgrade / downgrade / refer, with notes — saved to the record.

病例审阅：同意 / 升级 / 降级 / 转诊，可填写备注——写入档案。

4. Admin — Control the 3 API Keys • 管理员：管理 3 个 API 密钥

Log in as admin / 123456 → you land on /admin. Paste any of the 3 provider keys (ChatGPT / Gemini / Claude) and click Save. They apply to EVERY user instantly — web and the Telegram bot — no redeploy. Keys are masked after saving.

用 admin / 123456 登录后自动进入 /admin 页面。粘贴任意 3 个供应商密钥（ChatGPT / Gemini / Claude）并保存，立即对所有用户生效——包括网页端和 Telegram 机器人，无需重新部署。保存后密钥只显示掩码。

Key precedence everywhere: user's own pasted key → admin key → backend env key → offline mock.

密钥优先级（全局一致）：用户自己粘贴的密钥 → 管理员密钥 → 服务器环境变量密钥 → 离线 Mock。

5. Telegram Bot — Zero-Press • Telegram 机器人：零点击

Just send the bot a photo of the inside of a mouth — that's it. No buttons, no questions. The full agent pipeline runs immediately using the active demo patient's stored records (history, habits, last scaling date) and replies with BOTH reports plus the agent trace.

直接给机器人发送一张口腔内部照片即可——无需任何按钮或回答问题。系统立刻用当前演示患者的档案（病史、习惯、上次洗牙日期）运行完整智能体流水线，回复两类报告及智能体运行轨迹。

Command 指令	Effect 作用
(send a photo 发送照片)	Instant screening as the active demo patient 立即以当前演示患者身份筛查
/changepatient	Switch demo patient: Aisyah → Tan → Muthu 切换演示患者（别名 /anotherpatient)
/language	English ⇄ Bahasa Malaysia 切换语言
/help · /start	Show the welcome card with the active patient 显示欢迎信息与当前患者卡片

6. Two-Minute Demo Script • 两分钟演示脚本

- ① Login doctor/doctor → Patients → 'Load demo patients' → ask the triage agent who to see first.

① 用 doctor/doctor 登录 → 患者页 → 加载演示患者 → 问分诊智能体应先看谁。

- ② Login patient1 → New Screening → run a sample case → show the two-function result and the live agent pipeline animation.

② 用 patient1 登录 → 新筛查 → 运行示例病例 → 展示双功能结果与智能体流水线动画。

- ③ On Telegram, send a photo → instant result; /change patient → send again as high-risk Tan.

③ 在 Telegram 发一张照片 → 立即出结果; /change patient 后再发一张, 以高风险患者陈伟明身份筛查。

- ④ Open /introduction → walk the architecture diagram and the OpenClaw / Claude Code / Hermes pattern table.

④ 打开 /introduction → 讲解架构分层图, 以及借鉴 OpenClaw / Claude Code / Hermes 的设计模式表。

PART 2 • TECHNICAL REPORT — 技术报告

1. System Overview • 系统概览

OralScan AI is a hierarchical multi-agent system: ~14 specialised agents collaborate on every screening. One photo enters through any channel (web or Telegram); a single gateway routes it to the Orchestrator, which dispatches the specialist agents in order; every LLM output is structured JSON; every result becomes a shared record visible to both patient and doctor.

OralScan AI 是一个分层多智能体系统：每次筛查由约 14 个专职智能体协作完成。一张照片从任意渠道（网页或 Telegram）进入，由统一网关交给『编排者』智能体依序调度各专职智能体；所有大模型输出均为结构化 JSON；所有结果都写入共享存储，患者端与医生端同步可见。

2. Seven-Layer Architecture • 七层架构

Layer 层	What it does 职责	Key files 关键文件
1 · Channels 渠道	Patient web / Doctor web / Telegram / Admin — four doors, one brain 四个入口共用同一大脑	app/* · lib/telegram/handler.ts
2 · Gateway 网关	Role-checked auth, channel adapters, key resolution 角色鉴权、渠道适配、密钥解析	middleware.ts · app/api/*
3 · Lead agent 主智能体	Orchestrator dispatches all agents + audit log 编排者依序调度并记录审计日志	lib/agents/orchestratorAgent.ts
4 · Specialists 专职智能体	IoT → Vision → Risk → RAG → Expert Panel → Report → Referral → Triage → Dental 流水线	lib/agents/*
5 · Models 模型层	ChatGPT / Gemini / Claude / mock — swappable 可热切换	lib/visionProviders/*
6 · Memory 记忆层	KV records: sessions, profiles, messages, admin keys 会话、档案、消息、密钥记录	lib/server/repository.ts · kv.ts
7 · Outcomes 输出	Patient report · doctor queue · Telegram reply 三种视图同一数据	components/*

3. The Agent Registry • 智能体清单

Agent 智能体	Role 职责
Orchestrator 编排者	Knows the full pipeline; dispatches every agent; writes the audit log. 掌握全流程，逐一调度并记录审计日志。
Toothbrush IoT 牙刷物联网	Simulated smart-toothbrush telemetry: duration, pressure, coverage. 模拟智能牙刷遥测：时长、压力、覆盖率。
Vision Screening 视觉筛查	ONE image → TWO functions: cancer cues + tooth-health signs; provider pluggable with mock fallback. 一图双检：癌症线索 + 牙齿健康；模型可插拔并自动回退。
Cancer Risk Scoring 风险评分	Deterministic 0–100 score, literature-calibrated weights (IARC / INHANCE). 确定性 0–100 评分，权重源自文献。
Dental Wellness 牙齿健康	Cavity 蛀牙 / gums / plaque / staining / hygiene + scaling (洗牙) advice using the last-scaling record. 结合上次洗牙记录给出洗牙建议。

Agent 智能体	Role 职责
Multi-Expert Panel 专家会诊	3 role-prompted experts in PARALLEL (Pathologist on Claude; Epidemiologist + Dentist on Gemini) + deterministic Moderator. 三专家并行会诊 + 主持人汇总，保留分歧。
Consensus 共识核对	Second model re-reads the same image; agreement score. 第二个模型复核同一张图，计算一致性分数。
Retrieval (RAG) 知识检索	Grounds results in a curated oral-cancer knowledge base. 用口腔癌知识库为结论提供依据。
Patient Communication 患者沟通	Plain-language report with mandatory disclaimer. 通俗语言报告，强制附带免责声明。
Clinician Referral 转诊	Structured referral packet — generated ONLY on High risk. 仅在高风险时生成结构化转诊单。
Triage Prioritization 分诊排序	Urgency + SLA for the doctor queue. 为医生队列计算紧急度与时限。
Doctor Assistant 医生助手	REAL tool loop: model → tool → observation → repeat (≤4 steps) over a registry (profile / sessions / compare-all). 真实工具循环，最多 4 步，含全员比较工具。
Patient Chat 患者聊天	Multi-provider chain (Gemini → Claude → ChatGPT) + safe deterministic fallback. 多供应商链式回退 + 安全离线兜底。
Telegram Gateway 渠道网关	Zero-press channel adapter into the same pipeline and store. 零点击渠道适配器，与网页共用流水线与存储。

4. One Picture, Two Functions • 一图双检设计

A single shared prompt (lib/visionProviders/shared.ts) instructs whichever vision model is selected to return BOTH detections in one call: (1) oral-cancer cues — finding / region / probability / bounding boxes; (2) dental signs — cavity signs, gum redness, plaque/tartar, staining. The response is forced into typed JSON, coerced and clamped before any downstream agent reads it. The Dental Wellness agent then combines the image signs with toothbrush telemetry, the questionnaire, and the patient's last-scaling record to decide cavity risk and whether scaling is due.

统一提示词 (lib/visionProviders/shared.ts) 让所选视觉模型在一次调用中同时返回两类检测：（1）口腔癌线索——发现/部位/概率/检测框；（2）牙齿体征——蛀牙迹象、牙龈红肿、牙菌斑、染色。返回内容强制为类型化 JSON，并经过约束与裁剪后才进入后续智能体。『牙齿健康』智能体再结合牙刷遥测、问卷与上次洗牙记录，判断蛀牙风险与是否需要洗牙。

5. API Key Management • 密钥管理

Priority 优先级	Source 来源	Scope 作用范围
1 (highest 最高)	User-pasted demo key (browser localStorage) 用户自己粘贴的密钥（仅存浏览器）	That user's own requests 仅该用户的请求
2	Admin keys pasted at /admin (KV) 管理员在 /admin 粘贴的密钥（存 KV）	ALL users + Telegram, instantly 全体用户与 Telegram，即时生效
3	Backend env vars (Vercel) 服务器环境变量	Server-wide fallback 服务器级兜底
4 (lowest 最低)	Deterministic mock 离线确定性 Mock	Never crashes — degrades and says so 永不崩溃，降级并明确提示

Safety: keys are never logged or echoed; the admin API returns masked previews only; the Gemini key travels in a header (not the URL) to keep it out of proxy logs.

安全性：密钥不写日志、不回显；管理员接口仅返回掩码；Gemini 密钥放在请求头而非 URL，避免进入代理日志。

6. Patterns Borrowed from Leading Agent Harnesses • 借鉴的前沿智能体设计模式

Source 来源	Pattern 模式	Our implementation 我们的实现
OpenClaw	Gateway: many channels, one loop 多渠道单循环网关	Web + Telegram are thin adapters over one pipeline 网页与 Telegram 共用同一流水线
OpenClaw	Zero-press proactive agent 零点击主动智能体	Photo is the only input needed 一张照片即触发全流程
OpenClaw	Inspectable memory 可检视记忆	Plain JSON records per patient in KV 每位患者的纯 JSON 记录
Claude Code	Lead agent + parallel subagents 主智能体 + 并行子智能体	Orchestrator + 3-expert panel with moderator 编排者 + 三专家并行会诊
Claude Code	Structured outputs 结构化输出	Typed JSON everywhere; Gemini hard responseSchema 全链路类型化 JSON
Claude Code	Audit log + permissions 审计日志与权限	Per-session agent trace; role-gated APIs 每次会话的智能体轨迹；角色鉴权
Hermes	Self-describing tool registry 自描述工具注册表	doctorTools.ts — add a tool, agent uses it 新增工具即用
Hermes	Core loop: model→tool→observe→repeat 核心循环	Doctor Assistant ≤4-step tool loop with visible trace 可见工具轨迹
Hermes	Provider abstraction + fallbacks 模型抽象与回退	3 APIs behind one interface; graceful chain 三家 API 单接口，链式优雅降级

7. Technology Stack & Deployment • 技术栈与部署

Layer 层面	Technology 技术
Frontend 前端	Next.js 16 (App Router, Turbopack) · React 19 · TypeScript 5 · Tailwind CSS 3 · bilingual EN/BM UI 双语界面
Backend 后端	Next.js API routes (Node runtime) · Edge middleware auth · SSE streaming 实时流式进度
AI 模型	Gemini (generateContent + responseSchema) · Claude (Messages API) · ChatGPT (chat-completions, vision) · deterministic mock
Storage 存储	Upstash Redis (Vercel KV REST) · in-memory dev fallback 本地内存兜底
Channels 渠道	Web (Vercel) · Telegram webhook (production) / long-poll bot (dev)
Deployment 部署	GitHub → Vercel auto-deploy on push to main 推送即自动部署
Auth 鉴权	HMAC-SHA256 signed cookies (Web Crypto) · roles: patient / doctor / admin

8. Safety & Limitations • 安全与限制

- Educational prototype only — not a medical device; every output carries a disclaimer; the chat agent refuses diagnosis requests.

仅为教育原型——不是医疗器械；所有输出均附免责声明；聊天助手会拒绝诊断类请求。

- Demo-grade auth (seeded plaintext-compared passwords) and demo-grade key storage (KV, not a secret manager) — replace both before any real use.

演示级鉴权（内置明文比对密码）与演示级密钥存储（KV 而非专用密钥库）——任何真实使用前必须替换。

- Risk weights are literature-calibrated heuristics, not a clinically validated model; the toothbrush telemetry is simulated pending real IoT hardware.

风险权重为文献校准的启发式规则，并非经临床验证的模型；牙刷遥测目前为模拟数据，待接入真实硬件。

- If a provider key is missing or a call fails, the pipeline degrades to mock and clearly labels the fallback in the provider badge and audit log.

若缺少密钥或调用失败，流水线降级为 Mock，并在供应商标识与审计日志中明确标注。

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